Hof University
University of Applied Sciences

APPLIED DEEP LEARNING
00 – ORGANIZATIONAL MATTERS

PROF. DR. LARS ACKERMANN-IGL

A FEW WORDS ABOUT ME

2006-2009
Dipl.-Ing. | Trainee
IT
BERUFSAKADEMIE SACHSEN STAATLICHE STUDIENAKADEMIE DRESDEN UNIVERSITY OF COOPERATIVE EDUCATION
Transinsight GmbH

UNIVERSITÄT BAYREUTH
2010-2012
M. Sc.
Computer Science

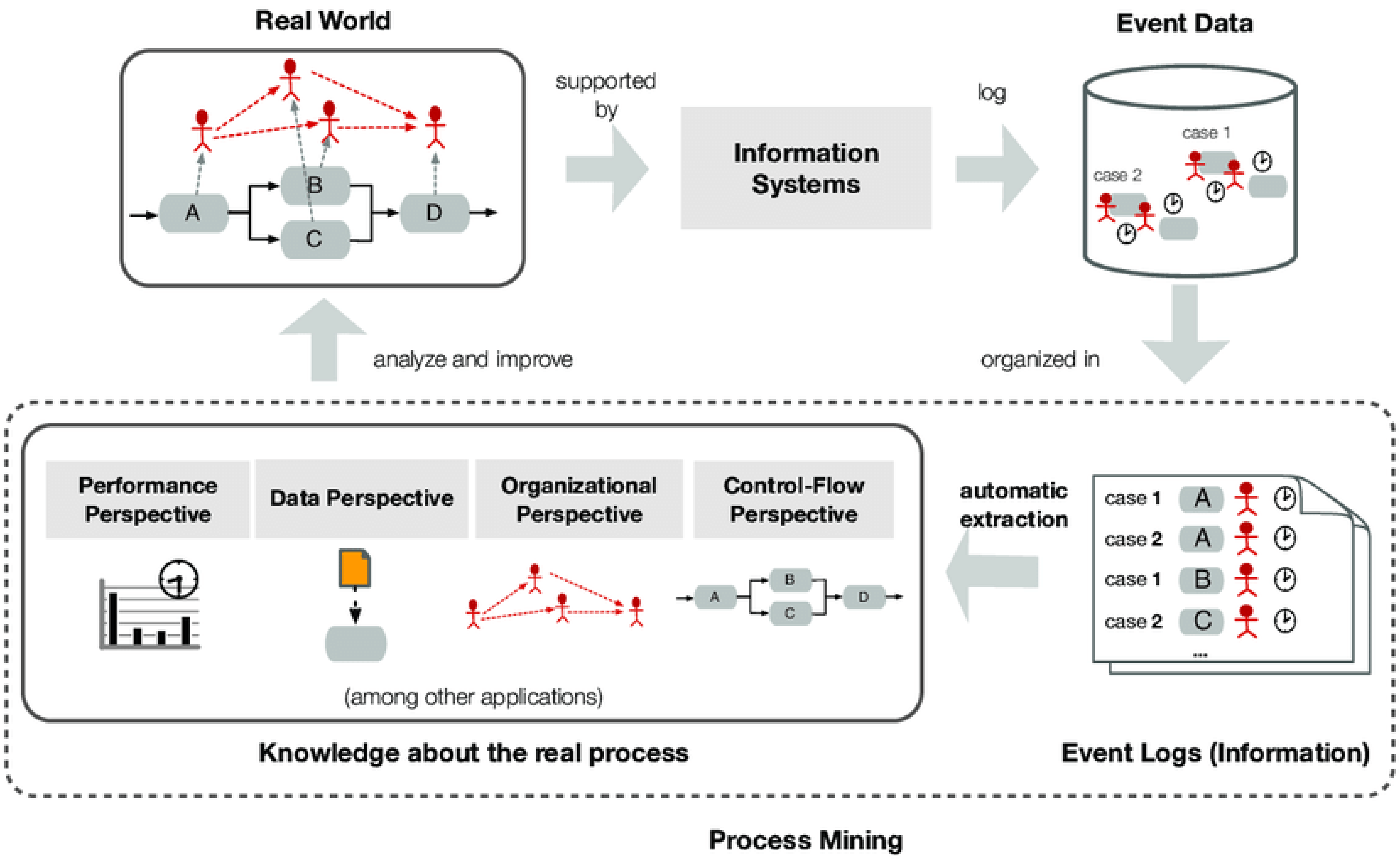
2013-2018
Databases and Information Systems
Prof. Dr.-Ing. Stefan Jablonski
PhD
Process Mining & Simulation
UNIVERSITÄT BAYREUTH

APPSALOT enterprise security
2018-today
Lead Software Architect
Founder
Process-driven Privacy

2018-01/2025
Databases and Information Systems
Prof. Dr.-Ing. Stefan Jablonski
Post doc
AI & Business Process Management
UNIVERSITÄT BAYREUTH

Hof University
02/2025-today
Process Mining
Professor
Unstructured data in Process Management

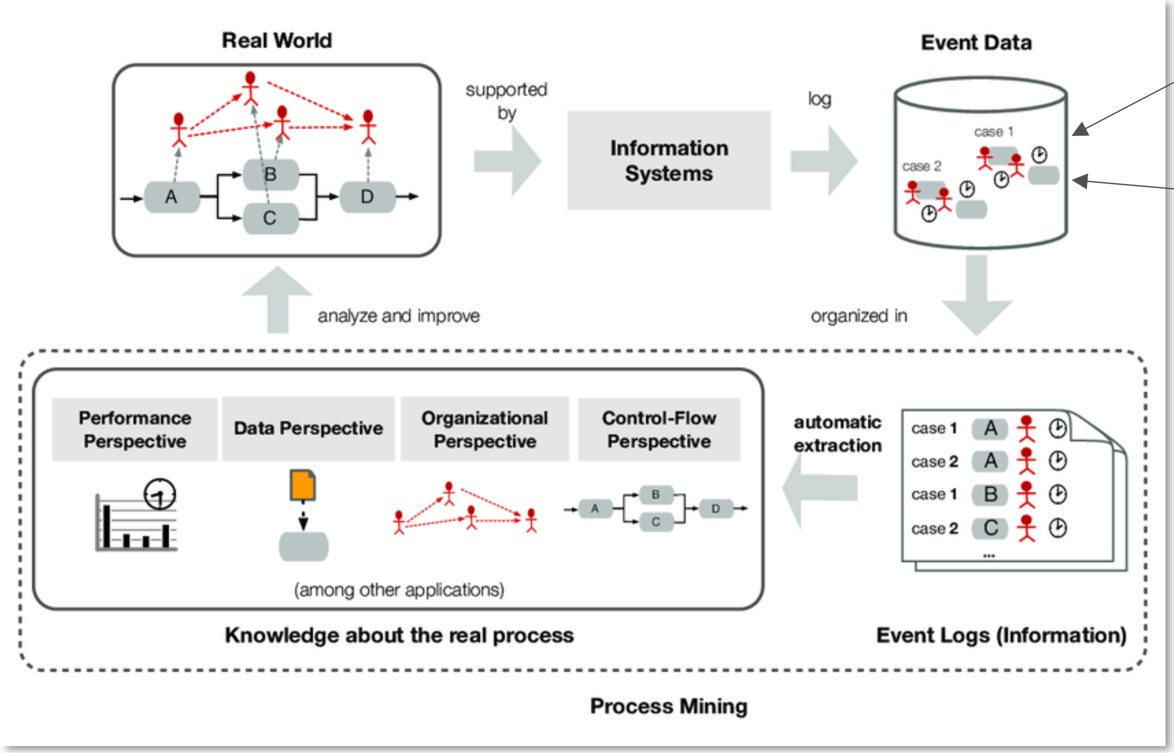
THE "PROCESS MINING RESEARCH GROUP" – WHAT IS (TRADITIONAL) PROCESS MINING?



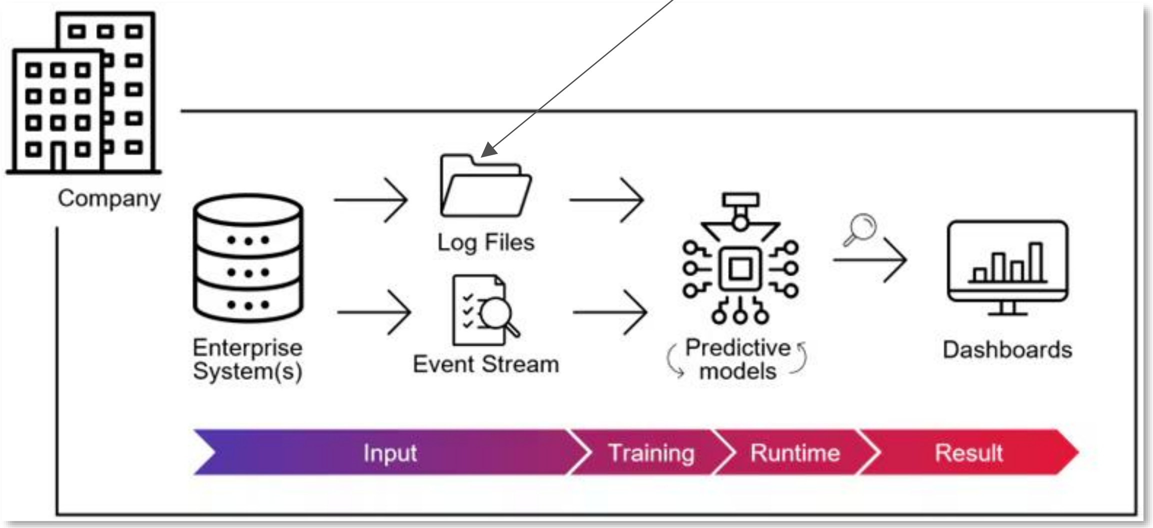
THE “PROCESS MINING RESEARCH GROUP” – OUR NATIVE RESEARCH FOCUS

Area 1: What can we do if there is no event data but process descriptions in natural language?

Area 2: How can we utilize unstructured data contained in event logs?



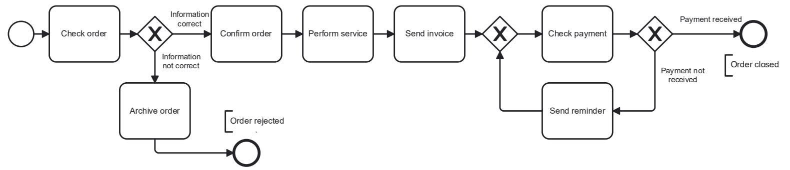
The traditional perspective



The (most popular) intersection with AI

THE “PROCESS MINING RESEARCH GROUP” – FURTHER RESEARCH INTERESTS – 1

- Large language models (GPT and similar modles for text generation, document search, automation, etc.)
- AI models for time series and sensor data
- AI models for event logs
- and much more



(Automatic) creation of formal, visual process models (from process data). Models can be used in particular for process control and automation.

- Determination of data characteristics (e.g., bias, distribution of inform. types)
- Review and, if necessary, improvement of data quality
- Data visualization (e.g., dashboards)

COMPETENCIES

Artificial Intelligence

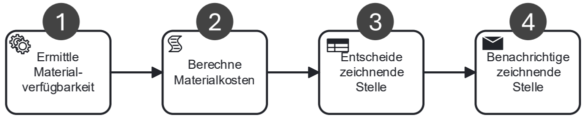
Process Modeling

Data Engineering

Process Automation

Natural Language Processing

Software Development



(1) Integration of external services and applications
(2) Customized automation
(3) Automated decisions and decision support (rules/AI)
(4) Automated email delivery or push notifications

- Text classification (e.g., by author, “hate speech”)
- Information extraction (e.g., locations and topics in social media posts)
- Chatbots for businesses and their documents

- AI prototypes
- Data analysis tools
- Web applications
- Middleware

MANY CONCEPTS ARE NOT ONLY RELEVANT IN PROCESS MINING—THERE ARE THEREFORE FURTHER POINTS OF CONNECTION.

THE "PROCESS MINING RESEARCH GROUP" – FURTHER RESEARCH INTERESTS – 2 (Coming soon)

Artificial Intelligence

Process Modeling

Data Engineering

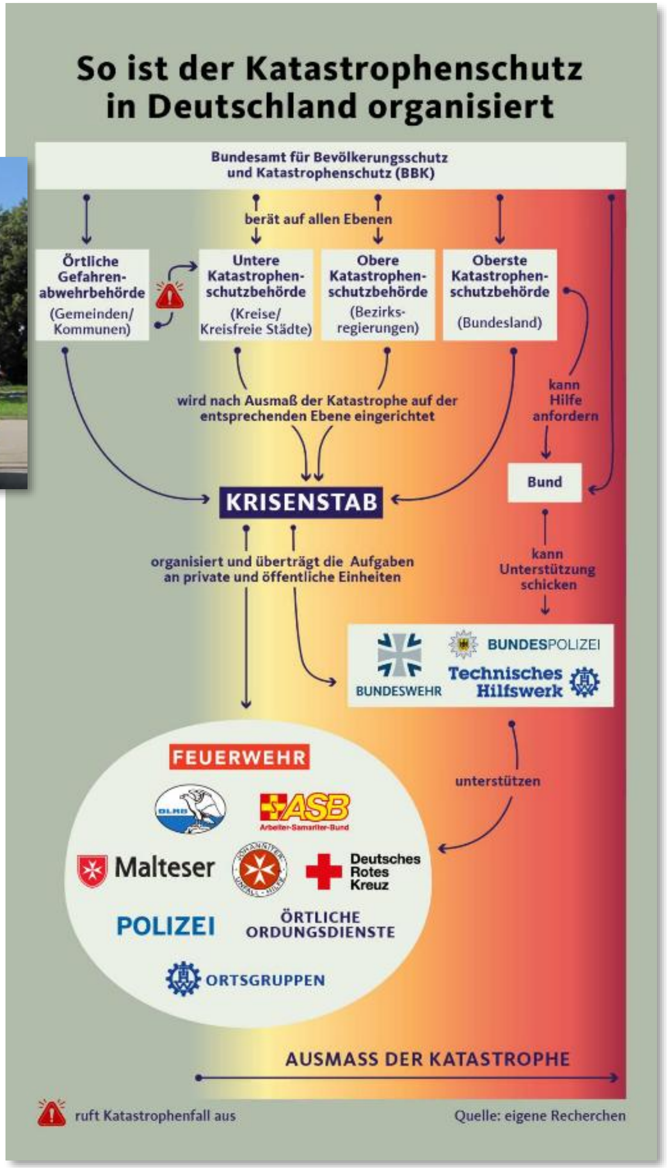
COMPETENCIES

Process Automation

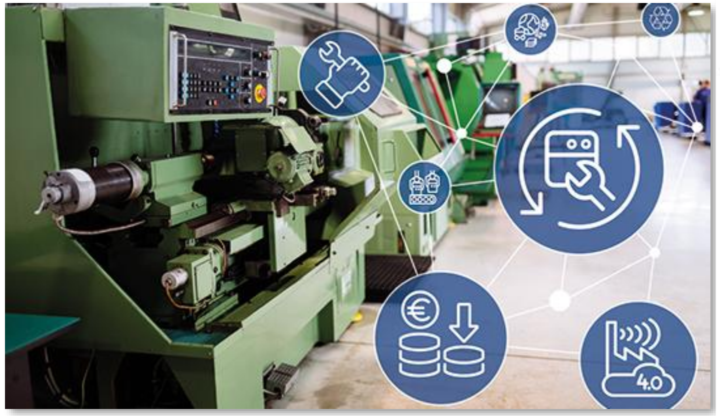
Natural Language Processing

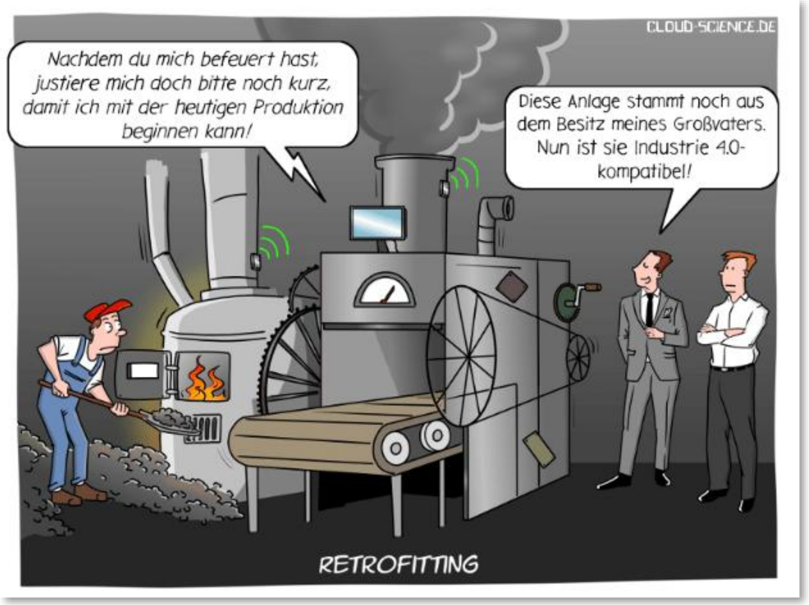
Software Development

Civil protection/ disaster management



Retrofitting





YOUR ASSOCIATIONS AND EXPECTATIONS

LET‘S DISCUSS YOUR EXPECTATIONS ☺

Poll Everywhere

Either via PollEv.com/larsackermann960 or QR code

5 Minutes

A FEW WORDS ABOUT THE COURSE

GOALS (according to the module manual)

Students know and understand the **structure** and **mode of action** of **deep neural networks**. They can prepare **data**, implement current **methods** using selected libraries and apply them to the data. They understand the effects of the individual parameters and can adjust them accordingly.

Course contents:

- Data Preparation (Splitting, Normalization, Standardization, Augmentation…)
- Introduction Neural Networks (Classification, Regression, Cost, Activation Functions)
- Deep Learning (Forward-, Backpropagation)
- Advanced Methods (Hyperparameter Tuning, Regularization, Learning Rate, Choice of Optimization Method)
- Convolutional Neural Networks (Convolution, Pooling Layer)
- Introduction to Deep Reinforcement Learning.

Note: We expect a diverse audience for this course, meaning that some of you will already have a solid background in AI and machine learning, while others will not. Therefore, the overall goal of this course is to bring everyone up to speed on the fundamentals of applied deep learning.

COURSE ORGANIZATION (PRELIMINARY)

CALENDAR WEEK	MONDAY	TOPIC
CW 12	☑	Organizational matters, Tools and Frameworks
CW 13	☑	Tools and Frameworks
CW 14	☑	
CW 15	Easter	Introduction to Machine Learning
CW 16	☑	Introduction to Machine Learning
CW 17	☑	Fundamentals of Neural Networks
CW 18	☑	Fundamentals of Neural Networks
CW 19	☑	Fundamentals of Neural Networks
CW 20	☑	
CW 21	☑	Convolutional Neural Networks
CW 22	Pentecost	Convolutional Neural Networks
CW 23	☑	
CW 24	☑	Reinforcement Learning
CW 25	☑	Reinforcement Learning
CW 26	☑	Efficient learning (HPO, Regularization, Accelerators)
CW 27	☑	Efficient learning (HPO, Regularization, Accelerators) | Buffer time
CW 28	☑	Buffer time / Q&A session

COURSE ORGANIZATION

Lecture sessions (room A009):
Monday 03:45 pm – 05:15 pm

Practical sessions (room B006):
Monday 11:30 am – 01:00 pm
Thursday 09:45 am – 11:15 am

EXAM

Written exam (90 minutes) at the end of semester, date tbd., no materials allowed

Exam period:
2026-07-11 – 2026-07-31

Changes and any further information will be announced in the Moodle course!

👉 ID 7325

HOW THE COURSE WORKS

LECTURE SESSIONS

- Presentation by lecturer: Learning concepts and their relationships
- Ad-hoc tasks (e.g., discussions, Jupyter notebooks): Hook for practical sessions

PRACTICAL SESSIONS

- Implementing concepts
- Exam preparation

It is recommended to solve all tasks in the practical session, too! Experience shows that success in exams directly correlates with exercise participation!

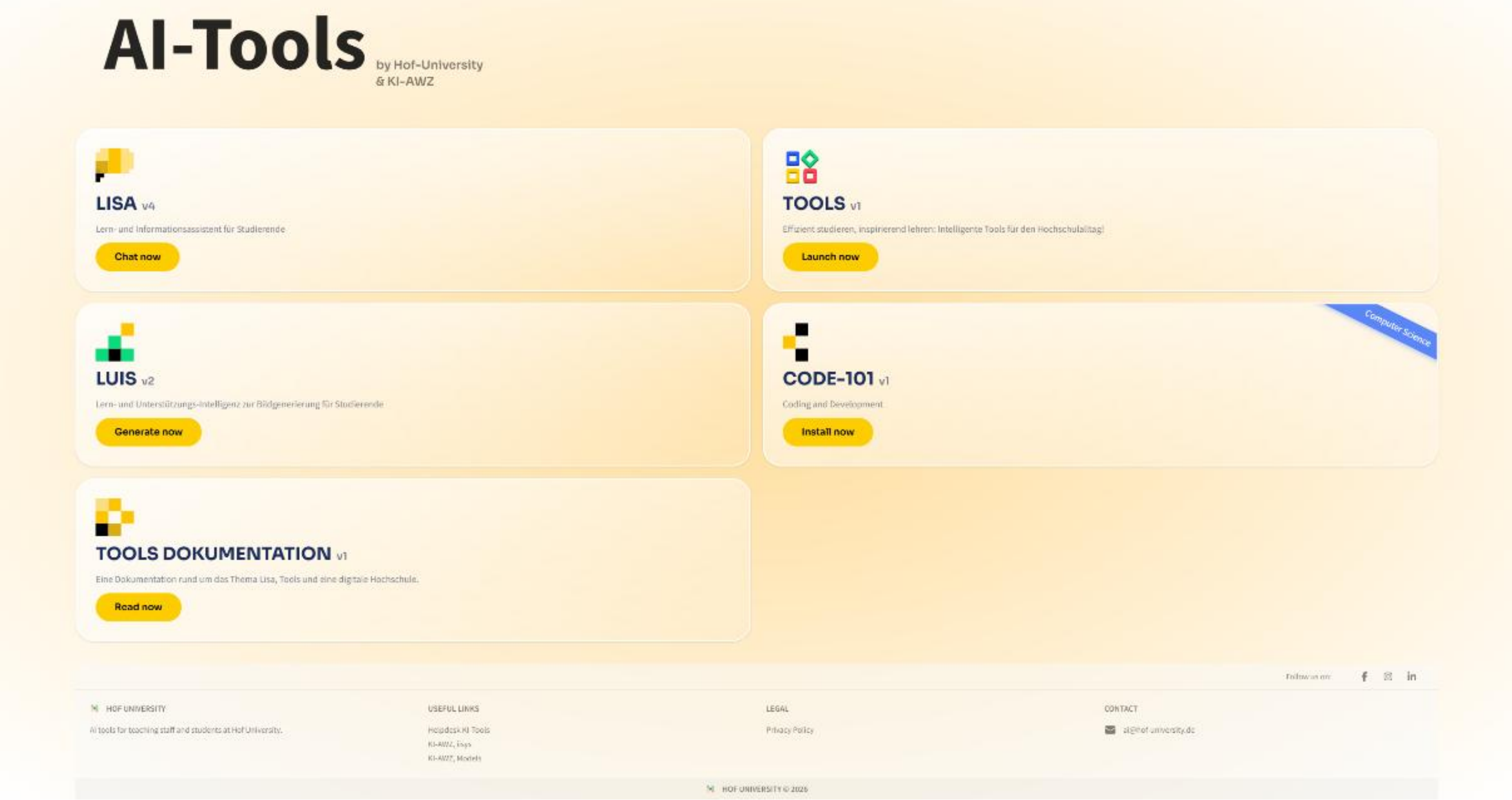
BETWEEN SESSIONS

- Self-study/exercise preparation or follow-up
- Ask questions: $channel = \begin{cases} Moodle\ course, & if\ also\ relevant\ for\ others \\ Teams\ (preferred)/Email/Appointment, & else \end{cases}$

“CAN WE USE GENERATIVE AI (IN EXERCISES)?”

Can I prevent it? ;-) Use it with caution!

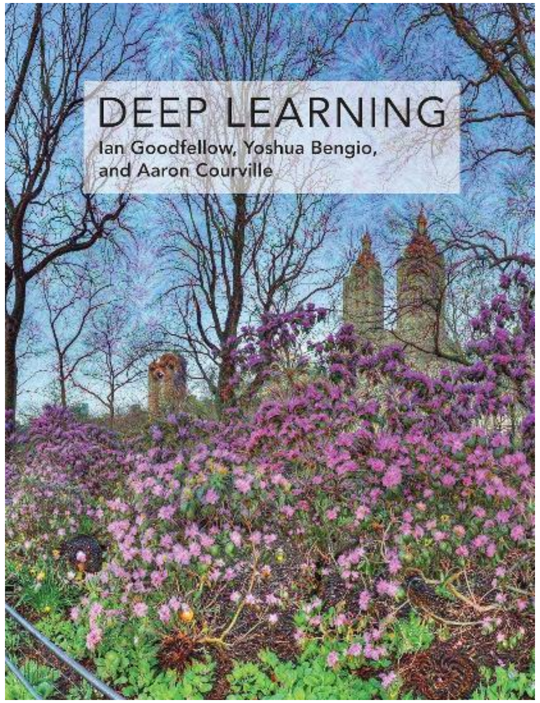
https://ai.hof-university.de/



AI does not prevent you from understanding, what you are doing. In the exam you won’t have access to an LLM! ;-)

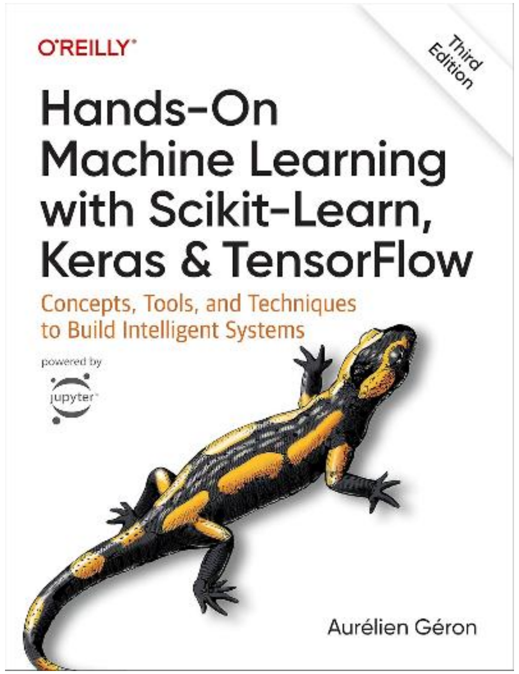
“CAN WE USE GENERATIVE AI?”

LITERATURE

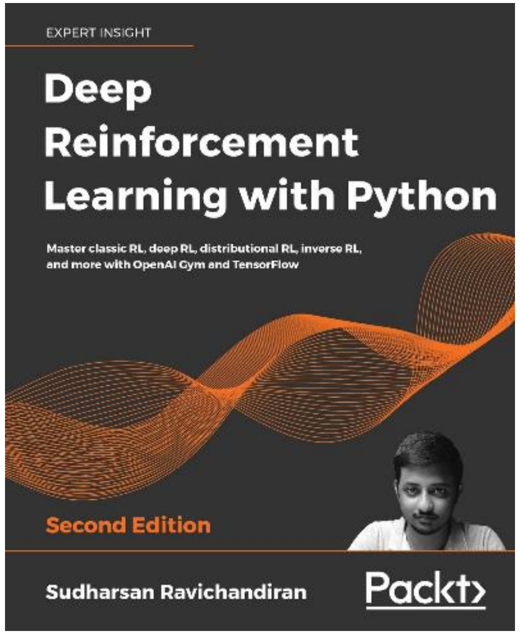


GOODFELLOW, Ian, et al. *Deep learning*. Cambridge: MIT press, 2016.

Online Version: https://www.deeplearningbook.org/



GÉRON, Aurélien. *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow*. " O'Reilly Media, Inc.", 2022.



RAVICHANDIRAN, Sudharsan. *Deep Reinforcement Learning with Python*. Birmingham, UK: Packt Publishing, 2020.

QUESTIONS?

DONE.

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